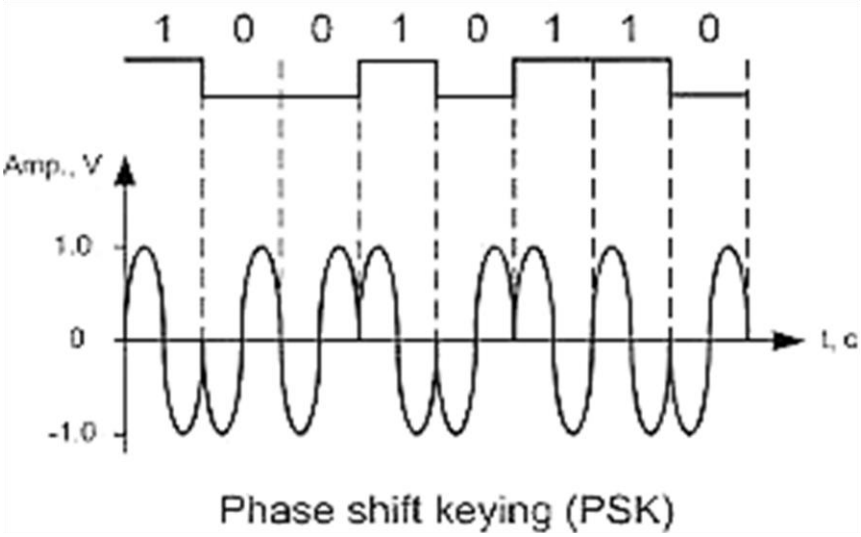
# A design of PSK Transmitter & Receiver Pair for Low Noise Transmission



Phase shift keying (PSK)

**EEE 310**

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

# EEE 310 (July 2023)

Communication Laboratory

# Final Project Report

## Section: C1 Group: 01

# A design of PSK Transmitter & Receiver Pair for Low Noise Transmission

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**Signature of Instructor:** _____

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## Contents

Abstract:....4
1. Introduction: ....4
2. Design: ....5
3. Description:....5
3.1 Encoding....6
3.1.1 ADC 0804:....6
3.1.2 SNx4HC165 8-Bit Parallel-Load Shift Registers....8
3.2 Modulation....11
3.3 Demodulation ....13
3.4 Decoding ....15
3.4.1 SNx4HC595: Serial in, parallel out ....15
3.4.2 DAC ....17
4. Result Analysis: ....20
5. Limitations: ....23
6. Conclusion & Future Work: ....23
7. References: ....24

# List of Illustrations

Figure 1 ADC 0804 ..... 6
Figure 2 Pin function ..... 7
Figure 3 Simulation diagram of ADC. ..... 7
Figure 4 Encoding Hardware Circuit ..... 8
Figure 5 Shift Resistor ..... 9
Figure 6 LF398 ..... 11
Figure 7 Modulation Circuit Simulation Diagram ..... 11
Figure 8 Modulation Hardware Circuit ..... 12
Figure 9: Functional Block Diagram of the Modulator ..... 12
Figure 10: PSK to ASK Conversion ..... 13
Figure 11 Demodulation Simulation Circuit ..... 14
Figure 12 Demodulation Hardware Circuit ..... 14
Figure 13 Decoder Circuit ..... 15
Figure 14 DAC Shift Registor ..... 16
Figure 15 Internal diagram of DAC ..... 18
Figure 16: Pin out of DAC0800 ..... 18

# Abstract:

**This project details the development of a Phase Shift Keying (PSK) transmitter and receiver system for transmitting analog signals over wired communication channels. The system employs an ADC0804 for analog to digital conversion, followed by PSK modulation and demodulation stages utilizing shift registers, op-amps, diode detectors, and comparators. Digital to analog conversion is achieved using a DAC0800 at the receiver end. Through successful implementation, this project demonstrates the viability of utilizing PSK modulation techniques for analog signal transmission, with potential applications in telecommunications, instrumentation, and control systems.**


# 1. Introduction:

In contemporary communication systems, the transmission of analog signals across wired communication channels continues to play a pivotal role in numerous applications spanning telecommunications, instrumentation, and control systems. Amidst the digital landscape, where data transmission primarily occurs in binary form, preserving the fidelity of analog signals necessitates effective modulation techniques. Phase Shift Keying (PSK) modulation stands out as an exemplary method for encoding analog information into digital signals, offering advantages such as robustness against noise and efficient bandwidth utilization.

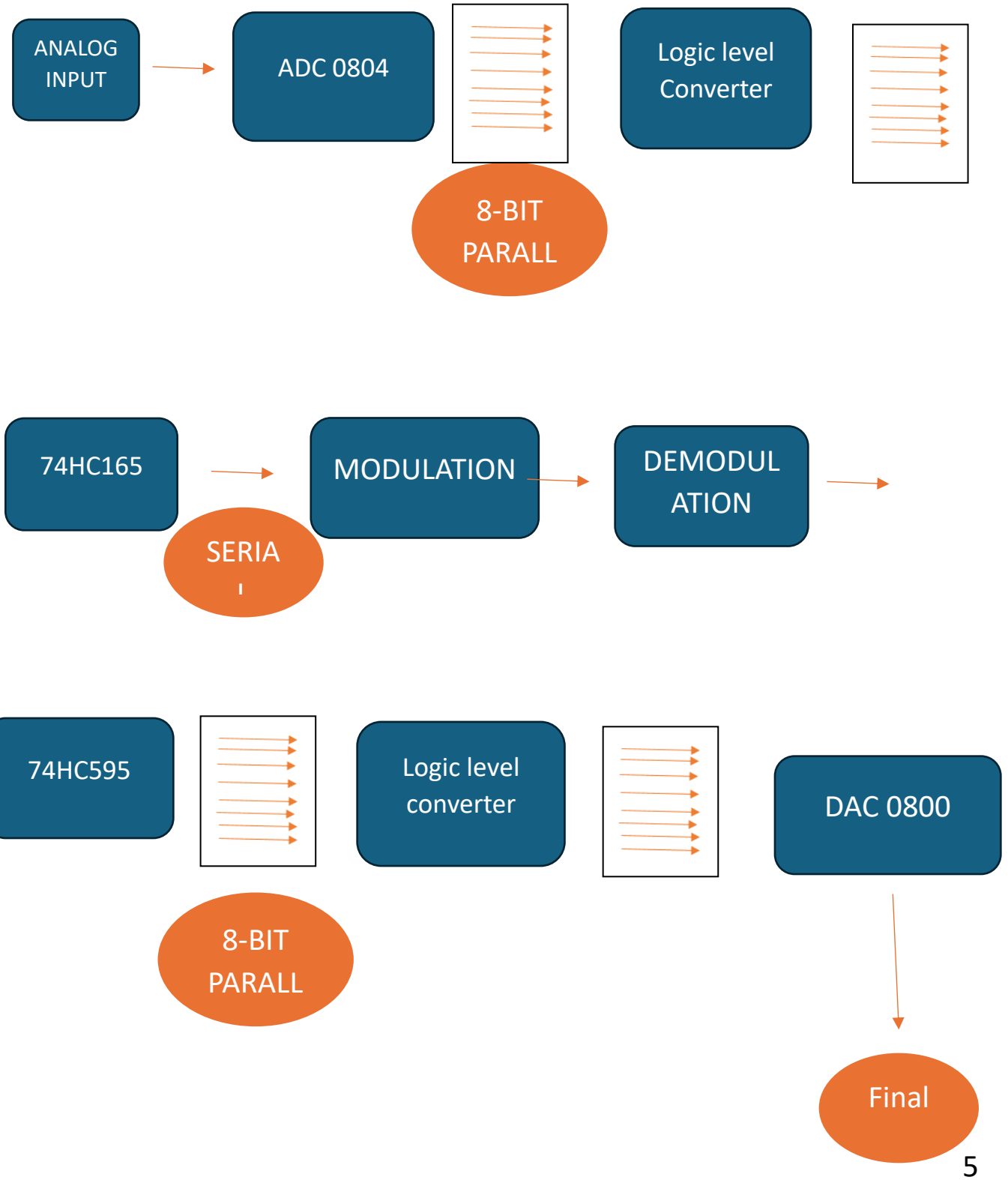
This project embarks on the development of a comprehensive PSK transmitter and receiver system tailored for the transmission of analog signals over wired mediums. The endeavor commences with the analog to digital conversion stage, facilitated by the implementation of the ADC0804. This crucial step enables the conversion of incoming analog signals into digital format, thereby facilitating subsequent processing. Following digitization, the digital signals undergo modulation onto carrier signals using PSK modulation techniques, orchestrated by a combination of components including shift registers, operational amplifiers, and the LF-398.

At the receiving end, the demodulation process takes center stage, facilitated by diode detectors and comparators. This step meticulously extracts the modulated analog information from the carrier signals, preparing it for the final stage of reconstruction. The serialized digital signals are subsequently deserialized using shift registers, paving the way for logic-level conversion and eventual reconstruction into analog form using the DAC0800.

The overarching objective of this project extends beyond mere implementation, aiming to validate the effectiveness and applicability of PSK modulation techniques in the context of analog signal transmission over wired communication channels. By showcasing the seamless integration of key components such as ADCs, shift registers, op-amps, diode detectors, comparators, and DACs, this project underscores the potential for PSK modulation to serve as a cornerstone in advancing the capabilities of wired communication systems. Through meticulous experimentation and analysis, this study seeks to contribute valuable insights to the broader discourse surrounding modulation techniques and their role in modern communication infrastructures.

## 2. Design:

Flow chart:



# 3. Description:

## 3.1 Encoding

Within our encoding framework, each step represents a meticulous orchestration of precision and ingenuity. The initial analog to digital conversion stage serves as the genesis, gracefully transforming the essence of analog signals into a digital canvas ripe for modulation. As the digital signals traverse the realm of modulation, they engage in a delicate dance of phase shifts, elegantly imbuing them with the essence of the carrier wave. Guided by the intricate interplay of components such as shift registers and operational amplifiers, the modulated signals transcend mere data, becoming vessels of nuanced expression. At the heart of our encoding journey lies the demodulation stage, a masterful unraveling of the modulated signals, revealing the hidden tapestry of analog information. With each meticulously crafted step, our encoding scheme emerges not just as a conduit of communication, but as a testament to the artistry inherent in the fusion of technology and creativity. Now we will explain the various components used in our encoding scheme.

### 3.1.1 ADC 0804:

PIN configuration:

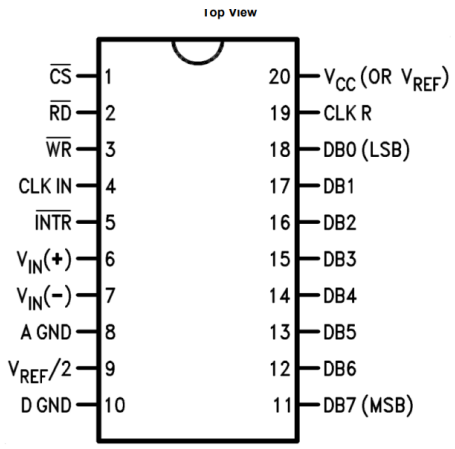



*Figure 1 ADC 0804*

**Pin Functions**

| PIN | | I/O | DESCRIPTION |
|---|---|---|---|
| NO. | NAME | | |
| 1 | CS | I | Chip Select |
| 2 | RD | I | Read |
| 3 | WR | I | Write |
| 4 | CLK IN | I | External Clock input or use internal clock gen with external RC elements |
| 5 | INTR | O | Interrupt request |
| 6 | $V_{IN}(+)$ | I | Differential analog input+ |
| 7 | $V_{IN}(-)$ | I | Differential analog input– |
| 8 | A GND | I | Analog ground pin |
| 9 | $V_{REF}/2$ | I | Reference voltage input for adjustment to correct full scale reading |
| 10 | D GND | I | Digital ground pin |
| 11 | DB7 | O | Data bit 7 |
| 12 | DB6 | O | Data bit 6 |
| 13 | DB5 | O | Data bit 5 |
| 14 | DB4 | O | Data bit 4 |
| 15 | DB3 | O | Data bit 3 |
| 16 | DB2 | O | Data bit 2 |
| 17 | DB1 | O | Data bit 1 |
| 18 | DB0 (LSB) | O | Data bit 0 |
| 19 | CLK R | I | RC timing resistor input pin for internal clock gen |
| 20 | $V_{CC}$ (or $V_{REF}$) | I | +5V supply voltage, also upper reference input to the ladder |

*Figure 2 Pin function*

The ADC0804 is an 8-bit analog-to-digital converter (ADC) integrated circuit (IC) that has been widely used in various electronic applications since its introduction. Manufactured by several semiconductor companies, including Texas Instruments, Analog Devices, and National Semiconductor, the ADC0804 offers a straightforward solution for converting analog signals into digital data. It operates on a successive approximation method, providing an 8-bit resolution which translates to 256 discrete voltage levels. The device typically features a moderate sampling rate, commonly around 6 to 8 kilo samples per second (kbps), making it suitable for applications where moderate speed is sufficient. The ADC0804 accepts analog input voltages in the range of 0 to 5 volts, although it can be interfaced with external circuitry to accommodate different voltage ranges if required. Its interface consists of several control lines, including Start, Clock, and Chip Select (CS), as well as an 8-bit parallel data output. The ADC0804 is designed to operate from a single power supply voltage, typically in the range of 4.5 to 6 volts, and it consumes relatively low power, making it suitable for battery-powered or low-power applications. While the ADC0804 provides reasonable accuracy and precision for many applications, its performance may be limited compared to more modern ADCs, particularly in terms of accuracy and speed. However, its simplicity, reliability, and affordability have made it a popular choice for hobbyists, students, and professionals alike in various applications, including data acquisition systems, instrumentation, process control, and digital audio equipment. Despite advancements in ADC technology, the

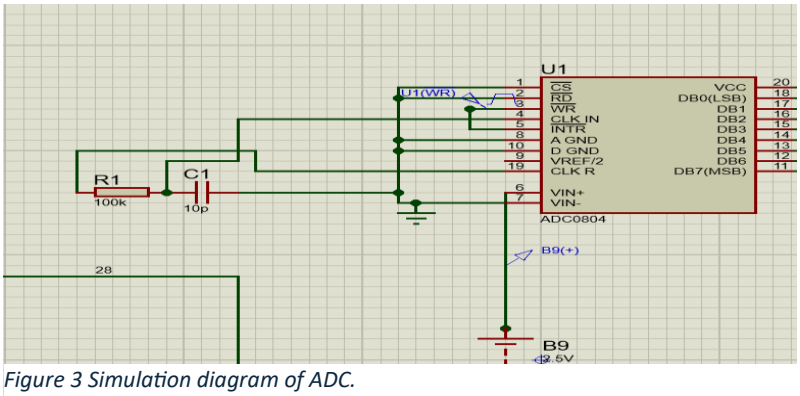



*Figure 3 Simulation diagram of ADC.*

ADC0804 continues to find utility in scenarios where its moderate specifications meet the requirements of the task at hand. The simulation schematics is given below.

The corresponding circuit block is shown below:

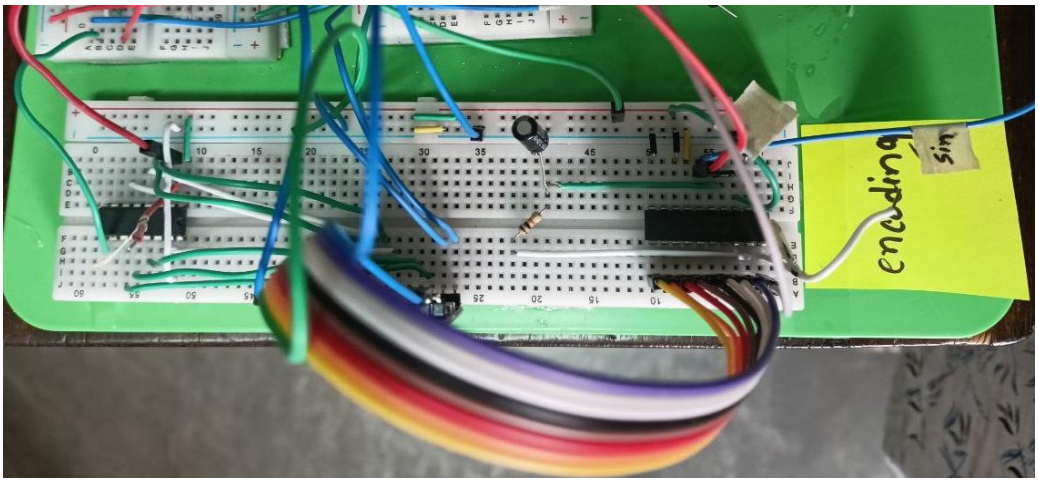



*Figure 4 Encoding Hardware Circuit*

### 3.1.2 SNx4HC165 8-Bit Parallel-Load Shift Registers

The SNx4HC165 devices are 8-bit parallel-load shift registers that, when clocked, shift the data toward a serial $(Q_H)$ output. Parallel-in access to each stage is provided by eight individual direct data $(A - H)$ puts that are enabled by a low level at the shift/load a clock-inhibit (CLK INH) function and a complementary serial $(\bar{Q}_H)$ output. Clocking is accomplished by a low-to-high transition of the clock (CLK) input while SH/LD is held high and CLK INH is held low. The functions of CLK and CLK INH are interchangeable. Because a low CLK and a low-to-high transition of CLK INH also accomplish clocking, CLK INH must be changed to the high level only while CLK is high. Parallel loading is inhibited when $\mathrm{SH}/\overline{\mathrm{LD}}$ is held high. While $\mathrm{SH}/\overline{\mathrm{LD}}$ is low, the parallel inputs to the register are enabled independently of the levels of the CLK, CLK INH, or serial (SER) inputs.

Features

- Wide Operating Voltage Range of 2 V to 6 V
- Outputs Can Drive Up to 10 LSTTL Loads

- Typical $t_{pd} = 13ns$
- $\pm$4-mA Output Drive at 5 V
- Low Input Current of 1$\mu$A Maximum
- Complementary Outputs
- Direct Overriding Load (Data) Inputs
- Gated Clock Inputs
- Parallel-to-Serial Data Conversion
- On Products Compliant to MIL-PRF-38535, All Parameters Are Tested Unless Otherwise Noted. On All Other Products, Production Processing Does Not Necessarily Include Testing of All Parameters.

Pin out of the IC:

**D, DB, N, NS, J, W, or PW Package**
**16-Pin SOIC, SSOP, PDIP, SO, CDIP, CFP, or TSSOP**
**Top View**

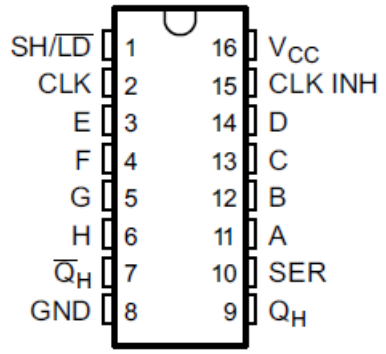



*Figure 5 Shift Resistor*

Pin Functions:

| PIN | | | I/O | DESCRIPTION |
|---|---|---|---|---|
| NAME | D, DB, N, NS, PW, J or W | FK | | |
| A | 11 | 14 | I | Parallel Input |
| B | 12 | 15 | I | Parallel Input |
| C | 13 | 17 | I | Parallel Input |
| CLK | 2 | 3 | I | Clock input |
| CLK INH | 15 | 19 | I | Clock Inhibit, when High No change in output |
| D | 14 | 18 | I | Parallel Input |
| E | 3 | 4 | I | Parallel Input |
| F | 4 | 5 | I | Parallel Input |
| G | 5 | 7 | I | Parallel Input |
| GND | 8 | 10 | — | Ground Pin |
| H | 6 | 8 | I | Parallel Input |
| NC | — | 1 | — | Not Connected |
| | | 6 | | |
| | | 11 | | |
| | | 16 | | |
| $Q_H$ | 9 | 12 | O | Serial Output |
| $\overline{Q}_H$ | 7 | 9 | O | Complementary Serial Output |
| SER | 10 | 13 | I | Serial Input |
| SH/$\overline{LD}$ | 1 | 2 | I | Shift or Load input, When High Data, shifted. When Low data is loaded from parallel inputs |
| $V_{CC}$ | 16 | 20 | — | Power Pin |

(1) NC – No internal connection

Truth Table:

| INPUTS | | | FUNCTION |
|---|---|---|---|
| SH/$\overline{LD}$ | CLK | CLK INH | |
| L | X | X | Parallel load |
| H | H | X | No change |
| H | X | H | No change |
| H | L | ↑ | Shift(1) |
| H | ↑ | L | Shift(1) |

(1) Shift : Content of each internal register shifts towards serial output $Q_H$. Data at SER is shifted into the first register

## 3.2 Modulation

For our modulator, we have used an LF 398 sample and hold circuit. The pinout of this IC is shown below:

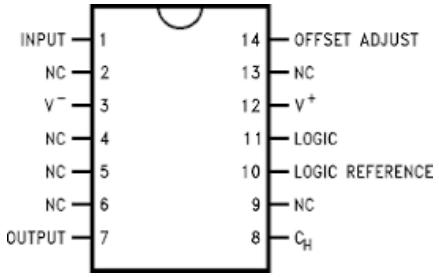



*Figure 6 LF398*

The functions of each pin is shown in the table below.

**Pin Functions**

| PIN | | | | TYPE(1) | DESCRIPTION |
|---|---|---|---|---|---|
| NAME | LF298, LF398-N SOIC-14 | LFx98x TO-99 | LF398-N PDIP-8 | | |
| $V^+$ | 12 | 1 | 1 | P | Positive supply |
| OFFSET ADJUST | 14 | 2 | 2 | A | DC offset compensation pin |
| INPUT | 1 | 3 | 3 | A | Analog Input |
| $V^-$ | 3 | 4 | 4 | P | Negative supply |
| OUTPUT | 7 | 5 | 5 | O | Output |
| $C_h$ | 8 | 6 | 6 | A | Hold capacitor |
| LOGIC REFERENCE | 10 | 7 | 7 | I | Reference for LOGIC input |
| LOGIC | 11 | 8 | 8 | I | Logic input for Sample and Hold modes |
| NC | 2, 4, 5, 6, 9, 13 | — | — | NA | No connect |

(1) P = Power, G = Ground, I = Input, O = Output, A = Analog

.

Digital input is given at pin 8 and analogue input is given at pin 3. The analogue input in this case is the carrier. At pin 6, an inverted version of this carrier is supplied. Schematics of the circuit is shown below:

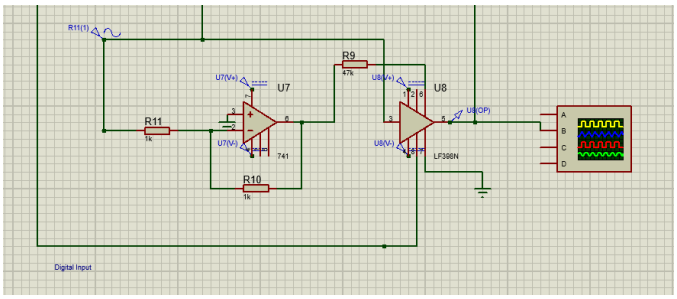



*Figure 7 Modulation Circuit Simulation Diagram*

The Corresponding Circuit Block is shown below:

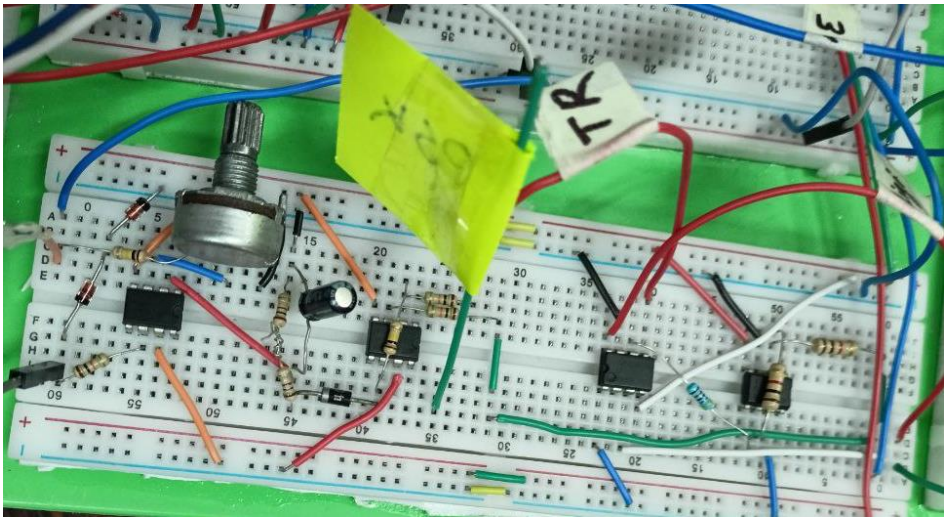


*Figure 8 Modulation Hardware Circuit*

Working Principle:

To understand the working of the modulator circuit, we need to see its functional block diagram. The figure below illustrates how the entire modulation process works.

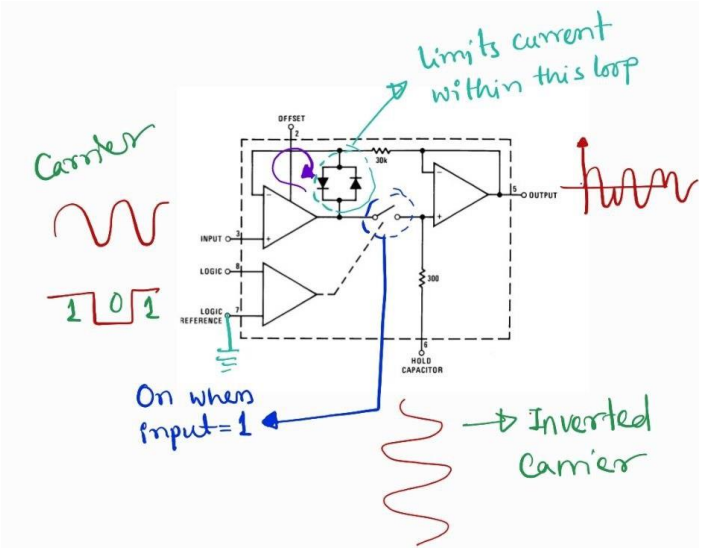



*Figure 9: Functional Block Diagram of the Modulator*

When the digital input at pin-8 is high, the switch in Fig-9 is closed and the output is connected to the analog input. Meaning, the output will be simply the analog input at this time. The intermediate circuit blocks act as voltage buffers.

When the digital input is low (as compared to reference), the switch is open and the output is disconnected from the input, and the signal at pin-6 is shown at the output during this time. An inverted version of the carrier is given at pin-6. So, whenever the digital signal makes a transition from 0 to 1 or 1 to 0, the output waveform seems to undergo 180 degrees phase shift. The result is the waveform drawn in red at the output pin-5, which is our desired modulated signal.

## 3.3 Demodulation

The demodulation task is done using envelop detection.

We know that PSK signals can't be detected non-coherently which is why we first converted our PSK signal into ASK by adding the carrier to it.

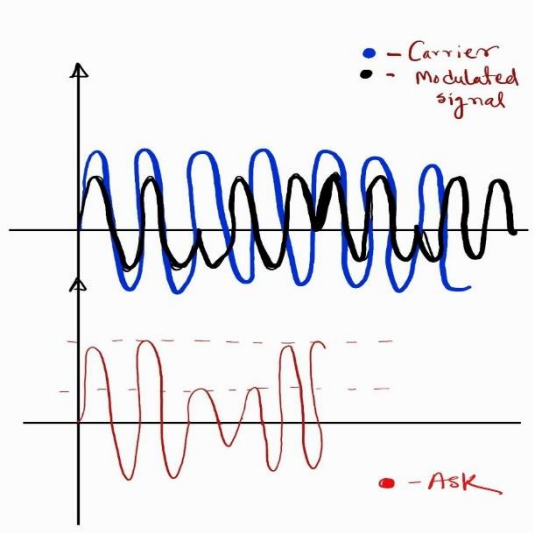



*Figure 10: PSK to ASK Conversion*

The ASK signal is then passed through a diode detector and comparator to get our original digital signal. The schematics of the circuit is shown below:

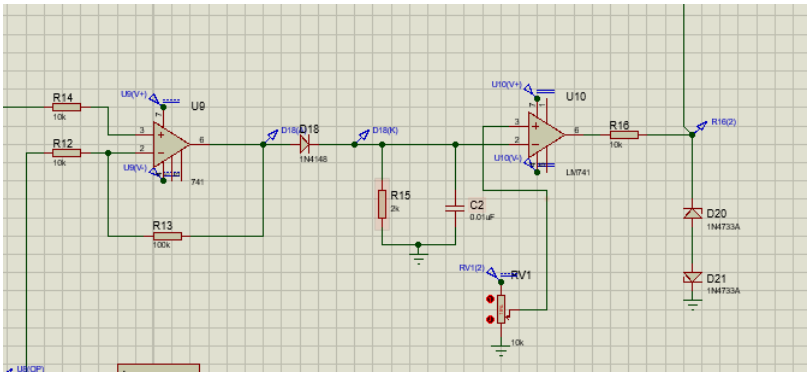



*Figure 11 Demodulation Simulation Circuit*

The corresponding circuit block is shown below:

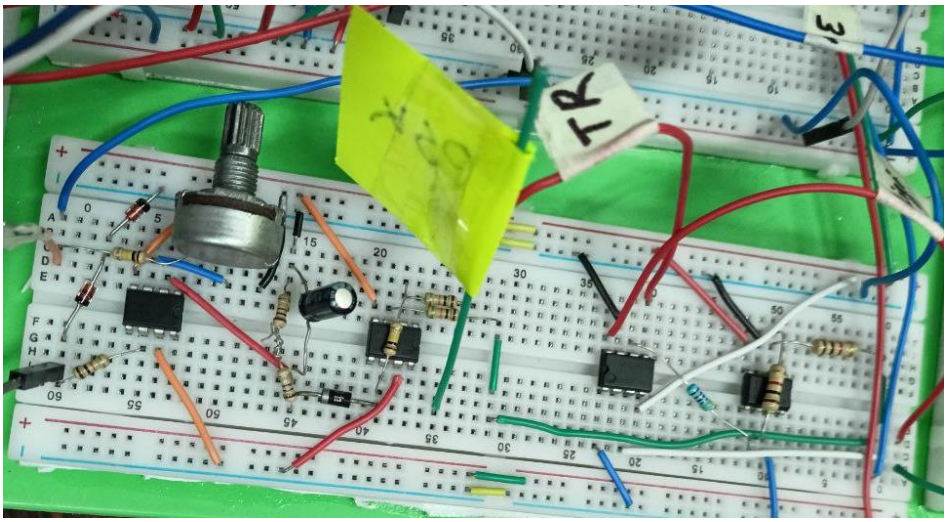


*Figure 12 Demodulation Hardware Circuit*

## 3.4 Decoding

The decoding process commences with the demodulation of the received signal using diode detectors, which extract the modulated analog information. The resulting analog signal is then refined through comparators, ensuring optimal signal quality and fidelity. Subsequently, the digital signal undergoes deserialization using shift registers, reverting it to its original parallel form for further processing or conversion back to analog. The corresponding circuit block is shown below:

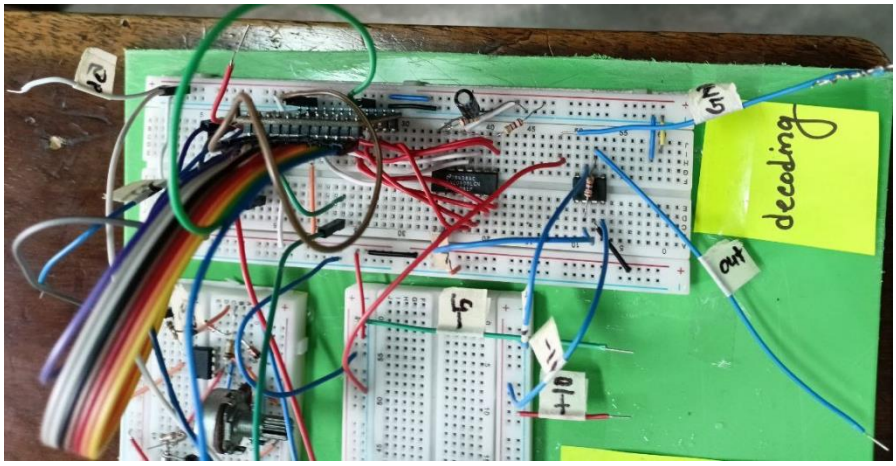



*Figure 13 Decoder Circuit*

### 3.4.1 SNx4HC595: Serial in, parallel out

The SNx4HC595 components feature an 8-bit shift register that takes in serial data and outputs it in parallel form. This feeds into an 8-bit storage register with parallel 3-state outputs. Each register has its own clock. The shift register has a clear input (SRCLR), serial input (SER), and serial outputs for cascading. When the output-enable (OE) input is set high, the outputs enter a high-impedance state.

**Features:**

- 8-bit serial-in, parallel-out shift register
- Wide operating voltage range: 2V to 6V
- High-current 3-state outputs capable of driving up to 15 LSTTL loads
- Low power consumption: maximum 80μA ICC
- Typical propagation delay (tpd): 13 ns
- ±6mA output drive at 5V
- Low input current: maximum 1μA
- Shift register features a direct clear

- For products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. For all other products, production processing may not include testing of all parameters.

Pin Configuration:

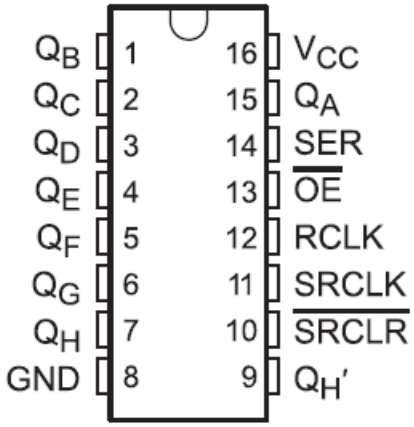



*Figure 14 DAC Shift Registor*

Pin Functions:

| PIN | | | I/O(1) | DESCRIPTION |
|---|---|---|---|---|
| NAME | SOIC, PDIP, SO, CDIP, SSOP, or TSSOP | LCCC | | |
| GND | 8 | 10 | — | Ground Pin |
| $\overline{OE}$ | 13 | 17 | I | Output Enable |
| $Q_A$ | 15 | 19 | O | $Q_A$ Output |
| $Q_B$ | 1 | 2 | O | $Q_B$ Output |
| $Q_C$ | 2 | 3 | O | $Q_C$ Output |
| $Q_D$ | 3 | 4 | O | $Q_D$ Output |
| $Q_E$ | 4 | 5 | O | $Q_E$ Output |
| $Q_F$ | 5 | 7 | O | $Q_F$ Output |
| $Q_G$ | 6 | 8 | O | $Q_G$ Output |
| $Q_H$ | 7 | 9 | O | $Q_H$ Output |
| $Q_{H'}$ | 9 | 12 | O | $Q_{H'}$ Output |
| RCLK | 12 | 14 | I | RCLK Input |
| SER | 14 | 18 | I | SER Input |
| SRCLK | 11 | 14 | I | SRCLK Input |
| $\overline{SRCLR}$ | 10 | 13 | I | $\overline{SRCLR}$ Input |
| NC | — | 1 | — | No Connection |
| | | 16 | | |
| | | 11 | | |
| | | 16 | | |
| $V_{CC}$ | — | 20 | — | Power Pin |

(1) Signal Types: I = Input, O = Output, I/O = Input or Output.

Functional Modes:

| INPUTS | | | | | FUNCTION |
|---|---|---|---|---|---|
| SER | SRCLK | $\overline{\text{SRCLR}}$ | RCLK | $\overline{\text{OE}}$ | |
| X | X | X | X | H | Outputs $Q_A$ – $Q_H$ are disabled. |
| X | X | X | X | L | Outputs $Q_A$ – $Q_H$ are enabled. |
| X | X | L | X | X | Shift register is cleared. |
| L | ↑ | H | X | X | First stage of the shift register goes low. Other stages store the data of previous stage, respectively. |
| H | ↑ | H | X | X | First stage of the shift register goes high. Other stages store the data of previous stage, respectively. |
| X | X | X | ↑ | X | Shift-register data is stored in the storage register. |

### 3.4.2 DAC

In general, it is known that a microprocessor generates only digital signal i.e., in the form of binary values as its output. However, in some applications like to control analog devices, analog signals are needed. Due to this reason, DAC interfacing is necessary as using a DAC the digital output of the microprocessor can be converted into analog form.

The input at the DAC unit is binary data which converts it into equivalent analog voltage or current. Every DAC contains 'n' number of input lines while a single output line. As we have discussed a reference value is required therefore, for the operation to take place DAC needs either a reference analog voltage ($V_{ref}$) or reference analog current ($I_{ref}$). A word resolution is used to describe the smallest analog value that an n-bit binary code can represent. For the n-bit binary value, DAC resolution is given as $1/2^n$ of the reference analog value. It is to be noted here that every analog output will be a multiple of the resolution.

Mainly the DAC is classified into 3 categories:

- Current Output DAC
- Voltage Output DAC
- Multiplying type DAC

As the name indicates, the current and voltage output DACs are the ones that generate analog current and voltage signals respectively as its output. The operation of voltage output DAC is such that internally the analog current signal is transformed into voltage equivalent. Other than these two types the third one is the multiplying type DAC that provides the output which is the product of applied input and the reference source. Though the difference is simple, even current output DAC and voltage output DAC can be considered as multiplying type DAC as all of them provide such an output which is the product of applied input with the reference value.

Block Diagram:

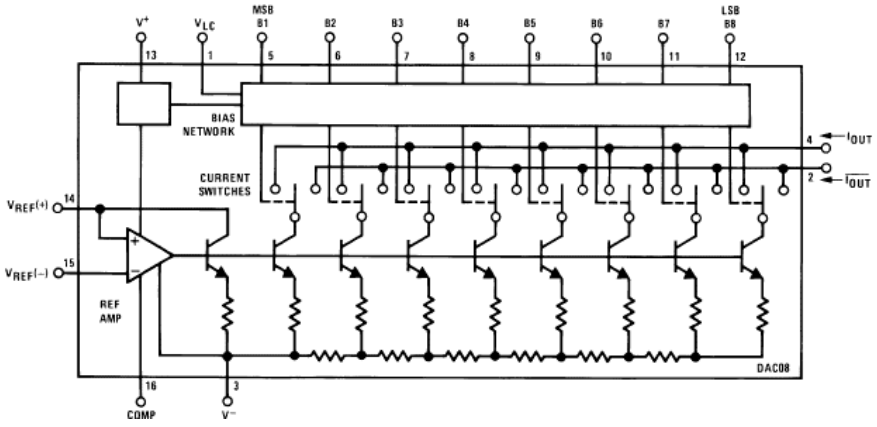



*Figure 15 Internal diagram of DAC*

Connection Diagram

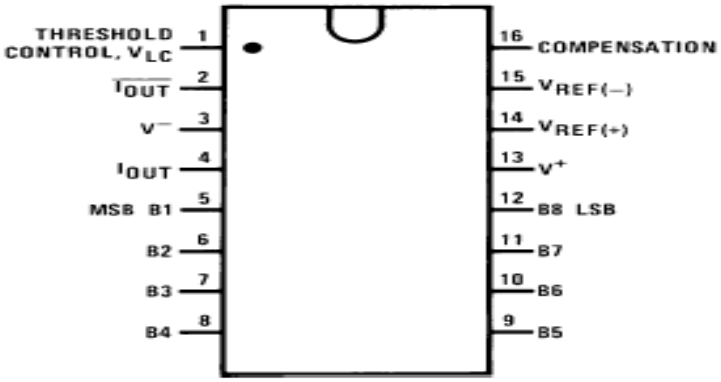



*Figure 16: Pin out of DAC0800*

As it is an 8-bit DAC it contains 8 digital input lines. Other than the input lines we have two dedicated pins that provide current output and complement of current output which is given the name $I_{OUT}$ and $I'_{OUT}$. There are a positive and a negative supply voltage pin along with a pin dedicated for compensation voltage. Here $V_{LC}$ represents the threshold control pin and there are two pins for positive and negative values of reference voltage denoted by $V_{REF}$ (+) and $V_{REF}$ (-).

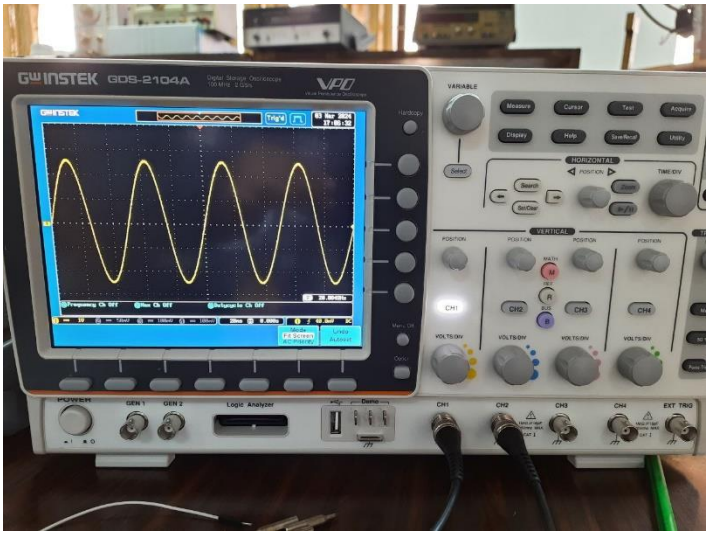
Absolute Maximum Ratings

| | |
|---|---|
| Supply Voltage ($V^+ - V^-$) | ±18V or 36V |
| Power Dissipation (2) | 500 mW |
| Reference Input Differential Voltage (V14 to V15) | $V^-$ to $V^+$ |
| Reference Input Common-Mode Range (V14, V15) | $V^-$ to $V^+$ |
| Reference Input Current | 5 mA |
| Logic Inputs | $V^-$ to $V^-$ plus 36V |
| Analog Current Outputs ($V_S-$ = −15V) | 4.25 mA |
| ESD Susceptibility (3) | TBD V |
| Storage Temperature | −65°C to +150°C |
| Lead Temp. (Soldering, 10 seconds) | |
| PDIP Package (plastic) | 260°C |
| CDIP Package (ceramic) | 300°C |
| Surface Mount Package | |
| Vapor Phase (60 seconds) | 215°C |
| Infrared (15 seconds) | 220°C |

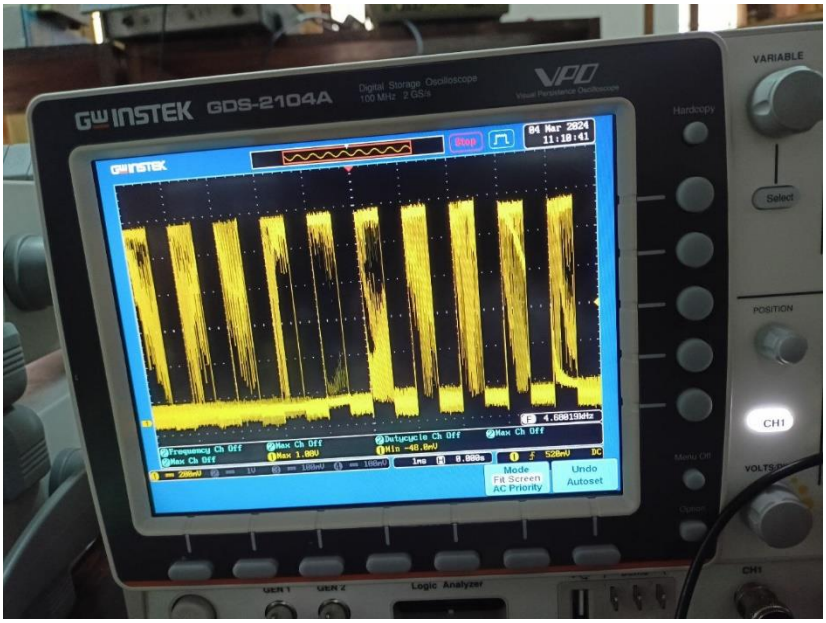
Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. DC and AC electrical specifications do not
apply when operating the device beyond its specified operating conditions.
(2) The maximum junction temperature of the DAC0800 and DAC0802 is 125°C. For operating at elevated temperatures, devices in the
CDIP package must be derated based on thermal resistance of 100°C/W, junction-to-ambient, 175°C/W for the molded PDIP package
and 100°C/W for the SOIC package.
(3) Human body model, 100 pF discharged through a 1.5 kΩ resistor.

# 4. Result Analysis:

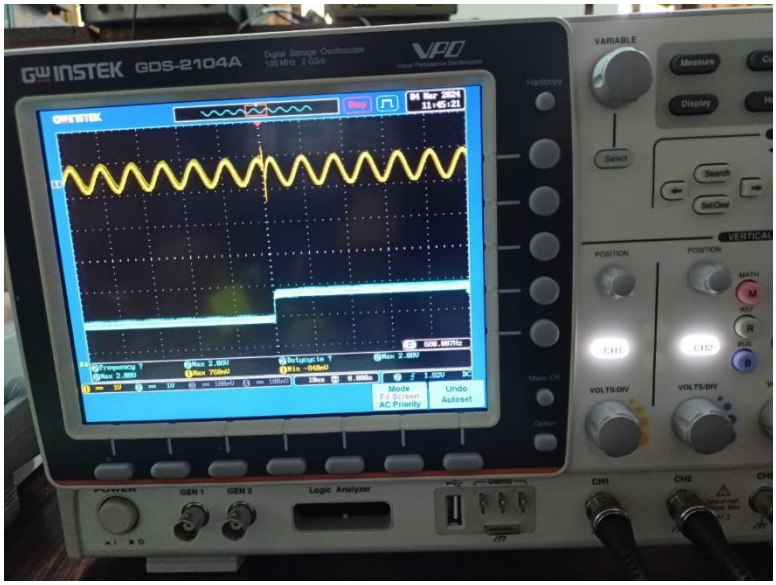
**For our analog message we have used sinusoid of 4V peak to peak and 2V offset. The oscilloscope waveform is shown below:**



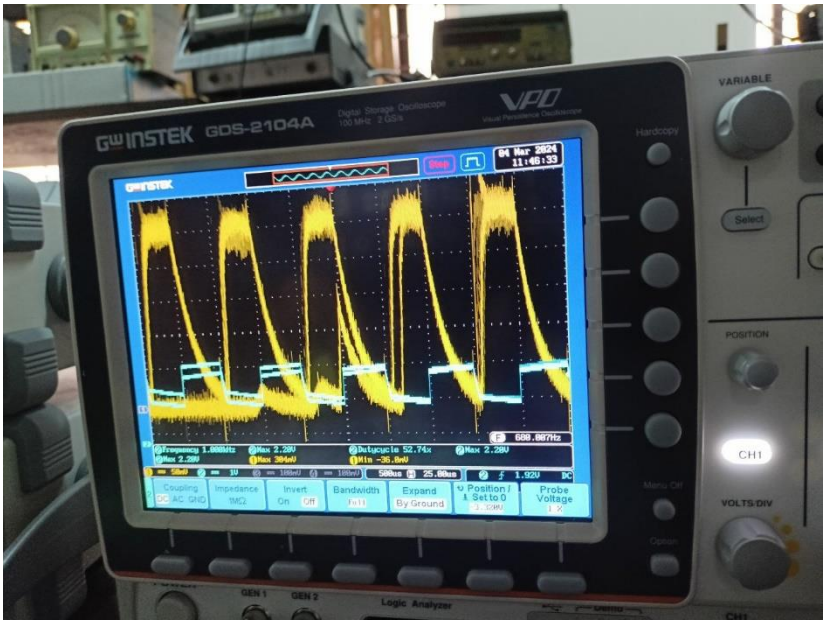

The ADC IC has 8 output pins. The waveform of one such pin is shown below:



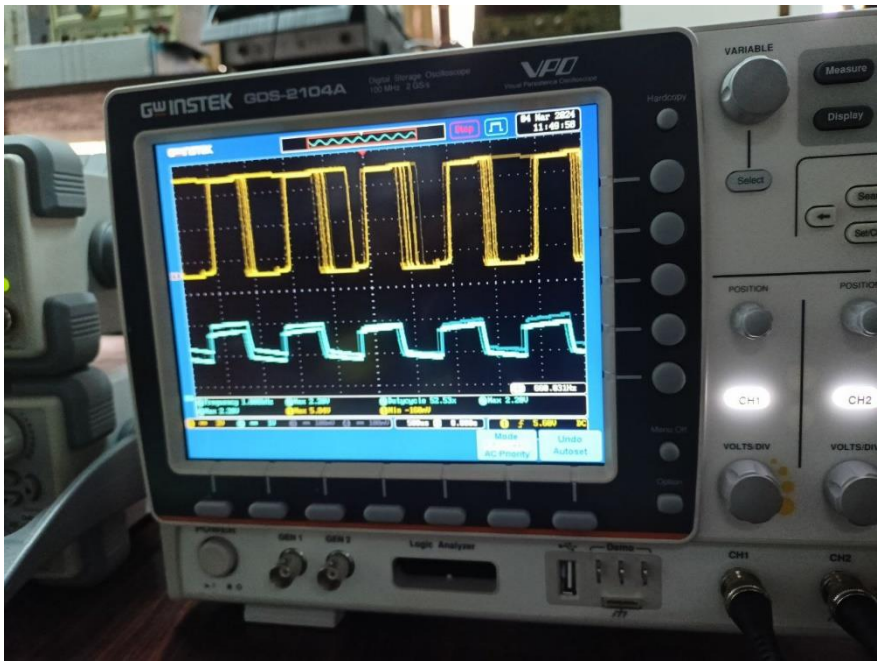
The modulated waveform was found to be the following figure.



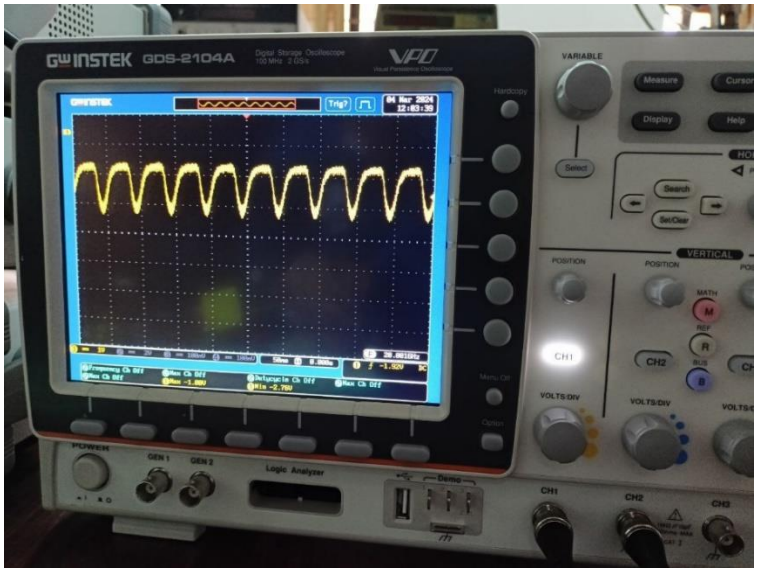

The waveform after crossing the diode detector is shown below.



The output from comparator is shown below.




The final reconstructed waveform was found to be somewhat like the following.



# 5. Limitations:

- The initial constraint lies in the utilization of ICs. The chosen ADC 0804 and DAC 0800 were not ideal for our intended application.
- Despite our preference for a different model, suitable ICs were unavailable.
- Three distinct clock frequencies were necessary, but timer circuits could not be developed due to the unavailability of required resistors and capacitors in the market.
- Attempts to implement oscillator circuits failed as the resistor and capacitor values in our circuit were tailored for a 128 kHz carrier frequency, which didn't align with our needs.
- Our chosen 8-bit linear encoding scheme was not optimal for voice and audio signal digitalization, where nonlinear encoding is typically more suitable.
- The bit error rate could have been mitigated with a higher bit depth, such as 16 bits or more, in the encoding scheme.

# 6. Conclusion & Future Work:

**Future Works:**

1. Enhanced Modulation Techniques: Explore advanced modulation techniques beyond PSK, such as Quadrature Amplitude Modulation (QAM) or Frequency Shift Keying (FSK), to ascertain their efficacy in enhancing data transmission rates and robustness.

2. Integration of Error Correction Codes: Investigate the integration of error correction codes, such as Reed-Solomon codes or convolutional codes, to fortify the system against channel impairments and enhance overall reliability.

3. Implementation of Adaptive Algorithms: Develop adaptive algorithms capable of dynamically adjusting modulation parameters based on channel conditions, thereby optimizing performance in varying environments.

4. Wireless Transmission: Extend the project scope to encompass wireless transmission, exploring the challenges and opportunities inherent in transitioning from wired to wireless communication mediums.

5. Real-Time Signal Processing: Investigate real-time signal processing techniques to streamline the encoding and decoding processes, minimizing latency and enhancing system responsiveness.

In conclusion, this project represents a significant stride towards leveraging sophisticated modulation techniques for the transmission of analog signals over wired communication channels.

Through the meticulous design and implementation of a Phase Shift Keying (PSK) transmitter and receiver pair, we have demonstrated the feasibility and efficacy of our encoding scheme in preserving signal integrity and enhancing noise immunity. As we look towards the future, the potential avenues for further exploration are abundant, ranging from advanced modulation techniques to adaptive algorithms and wireless transmission. With each step forward, we inch closer towards realizing the vision of seamless and efficient analog signal transmission, poised to revolutionize communication systems across diverse domains.

# 7. References:

**Shetty, M. (2014, October 23). Design of BPSK Modulator Using VHDL. INTERNATIONAL JOURNAL OF COMPUTERS & TECHNOLOGY, 13(12), 5247–5252. https://doi.org/10.24297/ijct.v13i12.5276**

**Datasheets**

**https://www.ti.com/lit/ds/symlink/sn74hc165**

**https://www.ti.com/lit/ds/symlink/adc0804-n**

**https://www.ti.com/lit/ds/symlink/sn74hc595**

**https://www.ti.com/lit/ds/symlink/dac0800**

**https://www.ti.com/product/LF398-N**